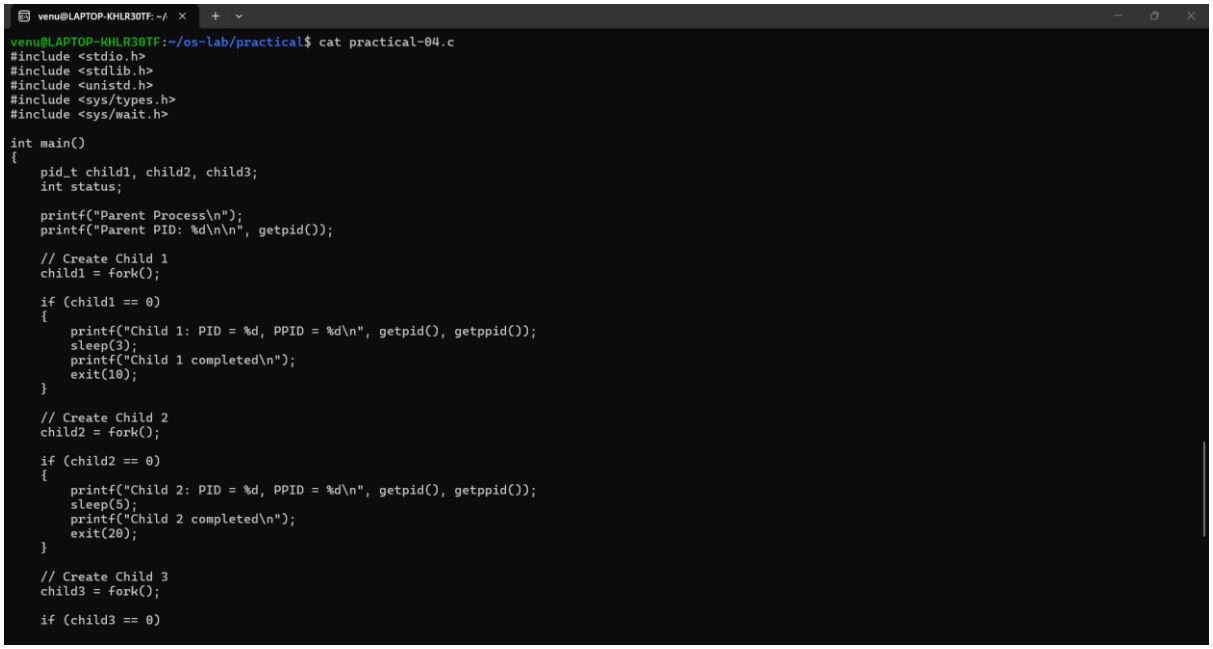
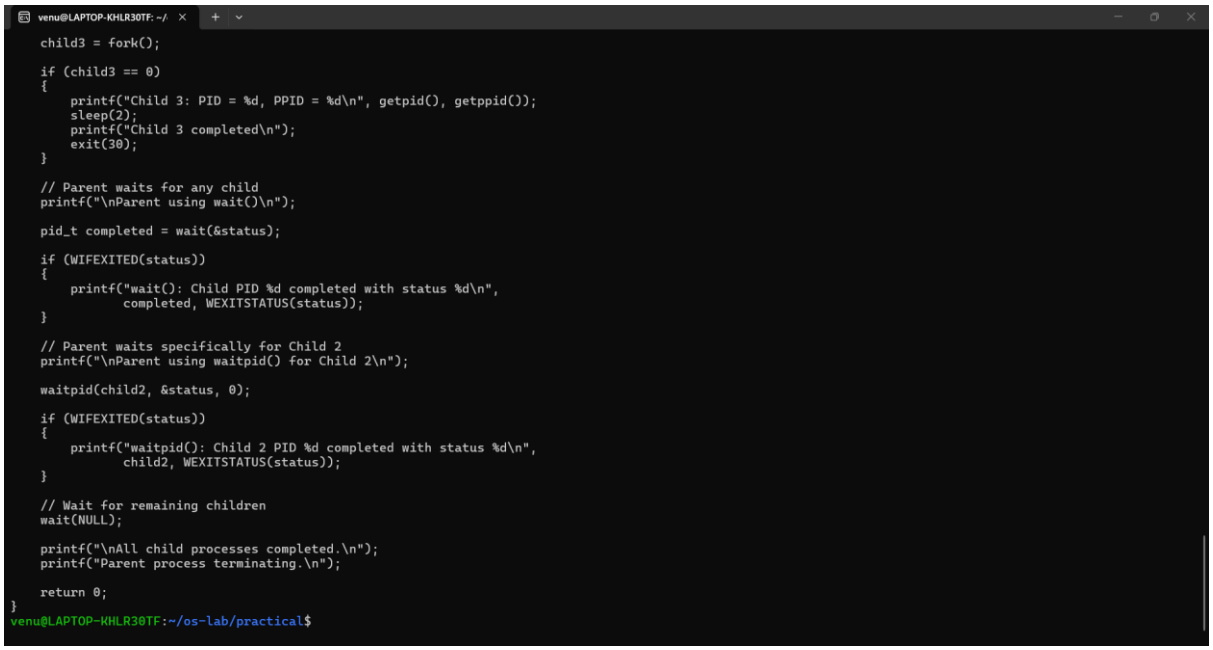
PRACTICAL-04

Write a C program where a parent process creates multiple child processes and synchronizes their completion using wait() and waitpid(). Compare the behavior of both functions.
Create a scenario where a child process becomes a zombie process. Investigate the process table and then modify the program to eliminate zombie processes using proper synchronization techniques.

CODE:-



```
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ cat practical-04.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main()
{
    pid_t child1, child2, child3;
    int status;

    printf("Parent Process\n");
    printf("Parent PID: %d\n\n", getpid());

    // Create Child 1
    child1 = fork();

    if (child1 == 0)
    {
        printf("Child 1: PID = %d, PPID = %d\n", getpid(), getppid());
        sleep(3);
        printf("Child 1 completed\n");
        exit(10);
    }

    // Create Child 2
    child2 = fork();

    if (child2 == 0)
    {
        printf("Child 2: PID = %d, PPID = %d\n", getpid(), getppid());
        sleep(5);
        printf("Child 2 completed\n");
        exit(20);
    }

    // Create Child 3
    child3 = fork();

    if (child3 == 0)
```



```
    child3 = fork();

    if (child3 == 0)
    {
        printf("Child 3: PID = %d, PPID = %d\n", getpid(), getppid());
        sleep(2);
        printf("Child 3 completed\n");
        exit(30);
    }

    // Parent waits for any child
    printf("\nParent using wait()\n");

    pid_t completed = wait(&status);

    if (WIFEXITED(status))
    {
        printf("wait(): Child PID %d completed with status %d\n",
               completed, WEXITSTATUS(status));
    }

    // Parent waits specifically for Child 2
    printf("\nParent using waitpid() for Child 2\n");

    waitpid(child2, &status, 0);

    if (WIFEXITED(status))
    {
        printf("waitpid(): Child 2 PID %d completed with status %d\n",
               child2, WEXITSTATUS(status));
    }

    // Wait for remaining children
    wait(NULL);

    printf("\nAll child processes completed.\n");
    printf("Parent process terminating.\n");

    return 0;
}
venu@LAPTOP-KHLR30TF:~/os-lab/practical$
```

OUTPUT:-

```
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ nano practical-04.c
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ gcc practical-04.c -o practical
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ ./practical-04
-bash: ./practical-04: No such file or directory
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ ./practical-04
-bash: ./practical-04: No such file or directory
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ gcc practical-04.c -o practical-04
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ ./practical-04
Parent Process
Parent PID: 956

Child 1: PID = 957, PPID = 956
Child 2: PID = 958, PPID = 956

Parent using wait()
Child 3: PID = 959, PPID = 956
Child 3 completed
wait(): Child PID 959 completed with status 30

Parent using waitpid() for Child 2
Child 1 completed
Child 2 completed
waitpid(): Child 2 PID 958 completed with status 20

All child processes completed.
Parent process terminating.
venu@LAPTOP-KHLR30TF:~/os-lab/practical$ cat practical-04.c
```